Final Practical: Program 1 to 6 with Code

Program 1: Classes and Objects in Java

Aim: To create a class and object in Java.

```
class Student {
    void display() {
        System.out.println("Hello Student");
    }

    public static void main(String[] args) {
        Student s = new Student();
        s.display();
    }
}
```

Output:

Hello Student

Program 2: Method and Constructor Overloading in Java

Aim: To demonstrate method and constructor overloading.

```
class Demo {
    Demo() {
        System.out.println("Default Constructor");
    }

    Demo(int a) {
        System.out.println(a);
    }

    void show(int a) {
        System.out.println(a);
    }

    void show(String s) {
        System.out.println(s);
    }

    public static void main(String[] args) {
```

```
        Demo d = new Demo();
        Demo d2 = new Demo(10);

        d.show(5);
        d.show("Hello");
    }
}
```

Output:

```
Default Constructor
10
5
Hello
```

Program 3: Inheritance and Exception Handling in Java

Aim: To demonstrate single inheritance and exception handling.

```
class A {
    void display() {
        System.out.println("Parent Class");
    }
}

class B extends A {
    public static void main(String[] args) {
        B obj = new B();
        obj.display();

        try {
            int x = 10 / 0;
        } catch (ArithmeticException e) {
            System.out.println("Error Handled");
        }
    }
}
```

Output:

```
Parent Class
Error Handled
```

Program 4: Create a Website Using HTML, CSS and JavaScript

Aim: To create a simple website using HTML, CSS and JavaScript.

```
<!DOCTYPE html>
<html>
<head>
   <title>My Website</title>
   <style>
      body {
         text-align: center;
         font-family: Arial;
      }

      h1 {
         color: blue;
      }
   </style>
</head>
<body>

   <h1>My Website</h1>
   <button onclick="show()">Click Me</button>

   <script>
      function show() {
         alert("Welcome to My Website");
      }
   </script>

</body>
</html>
```

Output: A web page displays a heading and a button. Clicking the button shows a welcome message.

Program 5: Email Validation Using JavaScript

Aim: To validate an email address by checking the presence of `@` and `.`.

```
<!DOCTYPE html>
<html>
<body>

   <h2>Email Validation</h2>

   <input type="text" id="email" placeholder="Enter Email">
   <button onclick="checkEmail()">Submit</button>

   <script>
      function checkEmail() {
```

```
        let e = document.getElementById("email").value;

        if (e.includes("@") && e.includes(".")) {
            alert("Email Accepted");
        } else {
            alert("Invalid Email. Enter Again");
        }
    }
</script>

</body>
</html>
```

Output: Displays an alert indicating whether the entered email contains both `@` and `.`.

Program 6: Change Background Color Every 5 Seconds Using DOM

Aim: To change the background color of a web page automatically every 5 seconds.

```
<!DOCTYPE html>
<html>
<body>

<h2>Background Color Change</h2>

<script>
    let colors = ["red", "blue", "green", "yellow"];
    let i = 0;

    setInterval(function() {
        document.body.style.backgroundColor = colors[i];
        i = (i + 1) % colors.length;
    }, 5000);
</script>

</body>
</html>
```

Output: The background color changes automatically between red, blue, green, and yellow every 5 seconds.

Practical execution steps

- Programs 1–3: Save each program as `Demo.java` or the matching public class filename, compile using `javac`, and run using `java`.

- Programs 4–6: Save each program in a separate `.html` file, such as `program4.html`, and open it in a web browser.